



TEST AND EXPERT REPORT

Correspondence: The petition by Zetaş Asansör İnşaat Pazarlama San. Tic. Ltd. Şti (Zetaş) registered on 10.04.2023.

Zetaş asked assessment about the nonwoven samples sent (labeled as S1 – S4) for the utilization of heat and sound isolation purposes. They had declared that the samples had keratin fiber (goat hair) and polyester (PET) staple fiber within. They also wanted testing study on various properties. So the report has assessment and test results parts

1. The Assessment

It is known that fibrous surfaces have been used as construction textiles in buildings for advanced mechanical properties and leakage, heat and sound insulation. Fiber reinforced building materials generally include synthetic fibers like glass, asbestos, polyester, polyamide, aramid thanks to high tensile properties, high elasticity modulus, round cross-section and hydrophobicity but natural fibers like cotton, flax, jute, coir and wool are also applicable (Gürdal and Acun, 2004, 2nd National Building Materials Congress and Exhibition Event, Kozak, 2013, SDÜ Natural and Technical Sciences Journal, 3,5, Türkmenoğlu and Varol, 2016, IMSEC).

Heat / sound insulation panels used in buildings are generally nonwoven felts. Nonwovens give sound absorption due to their micro-porous structure and large surface area and heat isolation due to thickness and air layer content. Those nonwoven panels are produced traditionally by web formation followed by mechanical or thermal bonding of virgin / recycled PET fibers. It is reported before that heat and sound isolation of those panels would be enhanced by increased thickness and weight; both decreased by increased porosity; on the other hand porosity and thermal conductivity of nonwovens would be changed by fiber blends (Yang et al., 2021, AUTEX Res. Journal., Yang et al. 2008, Textile Res. Journal, 89,4, Sakthivel et al., Journal of Eng. Fibers and Fabrics 2020, 15). Therefore it is possible to use staple fibers like cotton, jute and flax to increase the insulation performance of the panels. Also there are literature finding showing that synthetic and natural fiber blends would give better sound absorption than that of solely usage of those fibers in nonwoven based insulation panels (Sengubra, 2010, Indian J. Fibre Text. Research, 35, Sakthivel et al., 2020, Advances in Material Science and Eng., Sedlmajer et al., 2015,



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Adv. Mater. Res). There are commercially available insulation panels including waste cotton fibers (i.e. Echo Eliminator by Bonded Logic).

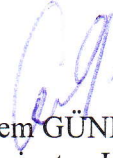
Goat hair is a keratin fiber with 18 – 25 μm fineness, 40 – 120 mm length, 2500 – 4500 aspect ratio, 15% moisture content, 90 – 120 MPa tensile stress (mid-high tensile fiber), and 30% elongation (elastic fiber) values. They are known with good abrasion resistance and higher cortex content than wool. Fine goat hair can be used for clothing whereas coarse fibers for outdoor clothing or insulation purposes. There are some studies on the successful usage of goat hair in concrete reinforcement even at low blend ratio as 0.8-1% giving advanced bending rigidity, compressive strength, water absorption and lowered spreading ratio (Şimşek and Kılıç, 2021, Journal of Innovations in Civil Engineering and Technology, 3, 2, Atav and Öktem, 2006, Tekstil ve Konfeksiyon, 6, Yarbaşı, 2020, Adıyaman University Journal of Engineering Sciences, 7, 13). These reports may encourage the utilization of goat hair in heat / sound insulation panels. Since goat hair, as a keratin fiber, has high heat of wetting, it would give good insulation even gets wet. The inherent nitrogen and sulfur content of the fiber would advance the flame retardant performance of the panels and their round cross-section would increase the adhesion to PET fiber giving better abrasion resistance.

2. The Test Results

The results of testing (average of four measurements) for the nonwoven samples sent by Zetaş are given in Table 1. The density values were obtained through weight and thickness measurement in accordance with ASTM D6242 and ASTM D5729 standards. The tensile strength measurements were completed in accordance with ISO 9073. Thermal conductivity measurements were completed according to the ALAMBETA SENSORA device method which satisfies the ISO 8301 standard. The air flow resistivity values were obtained through the measurement of air flow velocity through the specific area of the sample at 100 Pa pressure gradient in accordance with ASTM D 737 standard which satisfies ISO 9053 standard at final. The tests were run under standard laboratory conditions

Table 1. Test results

Sample	Weight (g/m ²)	Thickness (mm)	Density (kg/m ³)	Tensile Strength (N)	Thermal Conductivity (W/mK)	Air Flow Resistivity (kPa.s/m ²)
S1	400	2,5	160	520	0,034	104,62
S2	650	5,5	120	400	0,035	88,94
S3	850	7	120	900	0,036	87,34
S4	950	9,5	100	950	0,036	94,77
S5	1000	11,5	87	1000	0,039	100,55


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